Credit card customer segmentation

Unsupervised learning project

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Content

Define the problem and perform an Exploratory Data Analysis :- - Problem definition, questions to be answered - Data background and contents - Univariate analysis - Bivariate analysis - Insights based on EDA
Data preprocessing:- Prepare the data for analysis - Feature engineering - Missing value treatment - Outlier treatment - Duplicate observations check
Applying K-means Clustering:- - Apply K-means Clustering - Plot the Elbow curve - Check Silhouette Scores - Figure out appropriate number of clusters - Cluster Profiling
Applying Hierarchical Clustering:- - Apply Hierarchical clustering with different linkage methods - Plot dendrograms for each linkage method - Figure out appropriate number of clusters - Cluster Profiling
K-means vs Hierarchical Clustering:- Compare clusters obtained from K-means and Hierarchical clustering techniques
Actionable Insights & Recommendations:- Conclude with the key takeaways for the business - What would be your recommendations to the business?

Data Description :-

The data provided is of various customers of a bank and their financial attributes like credit limit, the total number of credit cards the customer has, and different channels through which customers have contacted the bank for any queries (including visiting the bank, online and through a call center).

Loading the dataset :-

	Sl_ No	Customer Key	Avg_Credit Limit	Total_Credit_ Cards	Total_visits_ bank	Total_visits_on line	Total_calls_ made
0	1	87073	100000	2	1	1	0
1	2	38414	50000	3	0	10	9
2	3	17341	50000	7	1	3	4
3	4	40496	30000	5	1	1	4
4	5	47437	100000	6	0	12	3
...	...	...	...	...	...	...	...
655	656	51108	99000	10	1	10	0
656	657	60732	84000	10	1	13	2
657	658	53834	145000	8	1	9	1
658	659	80655	172000	10	1	15	0
659	660	80150	167000	9	0	12	2

- There are 660 records, and 7 columns in the dataset

  - All the columns in the data are numeric.

  - There are no null value present in the dataset.

Sl_No	0
Customer_Key	0
Avg_Credit_Limit	0
Total_Credit_Cards	0
Total_visits_bank	0
Total_visits_online	0
Total_calls_made	0
dtype: int64	

  - Checking the unique values present in the dataset.

Sl_No	660
Customer Key	655
Avg_Credit_Limit	110
Total_Credit_Cards	10
Total_visits_bank	6
Total_visits_online	16
Total_calls_made	11
dtype: int64	

  - Checking for duplicate values present in the dataset.

	Sl_No	Customer Key	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
48	49	37252	6000	4	0	2	8
432	433	37252	59000	6	2	1	2

	Sl_No	Customer Key	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
4	5	47437	100000	6	0	12	3
332	333	47437	17000	7	3	1	0

	Sl_No	Customer Key	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
411	412	50706	44000	4	5	0	2
541	542	50706	60000	7	5	2	2

	Sl_No	Customer Key	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
391	392	96929	13000	4	5	0	0
398	399	96929	67000	6	2	2	2

	Sl_No	Customer Key	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
104	105	97935	17000	2	1	2	10
632	633	97935	187000	7	1	7	0

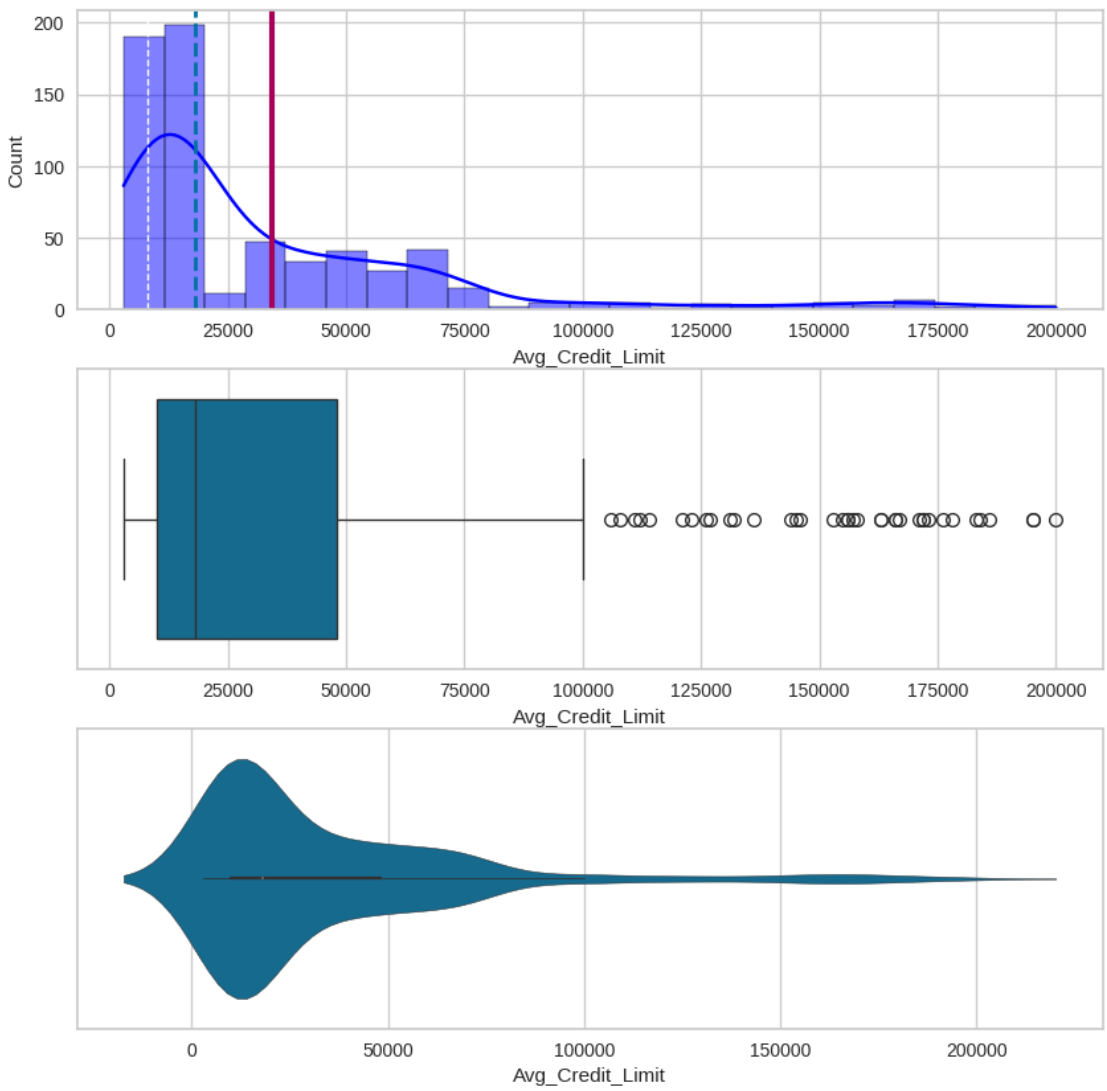
- Statistical summary of the dataset

	count	mean	std	min	25%	50%	75%	max
Avg_Credit_Limit	660.0	34574.242424	37625.487804	3000.0	10000.0	18000.0	48000.0	200000.0
Total_Credit_Cards	660.0	4.706061	2.167835	1.0	3.0	5.0	6.0	10.0
Total_visits_bank	660.0	2.403030	1.631813	0.0	1.0	2.0	4.0	5.0
Total_visits_online	660.0	2.606061	2.935724	0.0	1.0	2.0	4.0	15.0
Total_calls_made	660.0	3.583333	2.865317	0.0	1.0	3.0	5.0	10.0

EDA

Univariate analysis

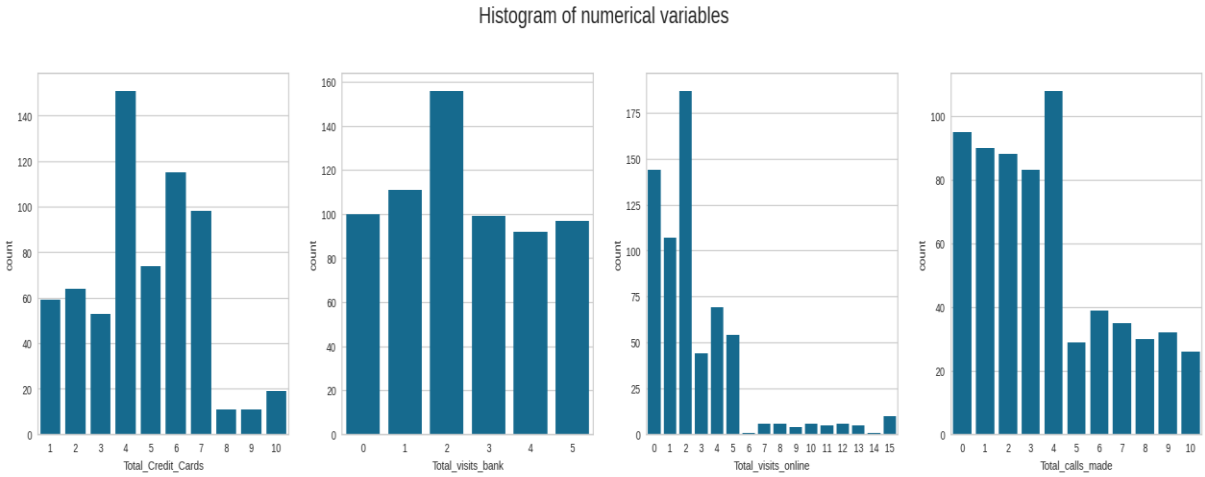
Avg_Credit_Limit



Observations:

- The Avg_Credit_Limit of a customer varies from 3000 to 200000.
- Avg_Credit_Limit is highly righly skewed.
- There are lot of outliers to the right.
- The mean of Avg_Credit_Limit is 34574 while the median is 18000.
- The 3rd quantile value is at 48000 while the first quantile value is at 10000 and the median is at 18000.
- Majority of the customers have Avg_Credit_Limit lesser than the median value-18000.
- There are quite some customers whose average credit limit is higher than 100000.

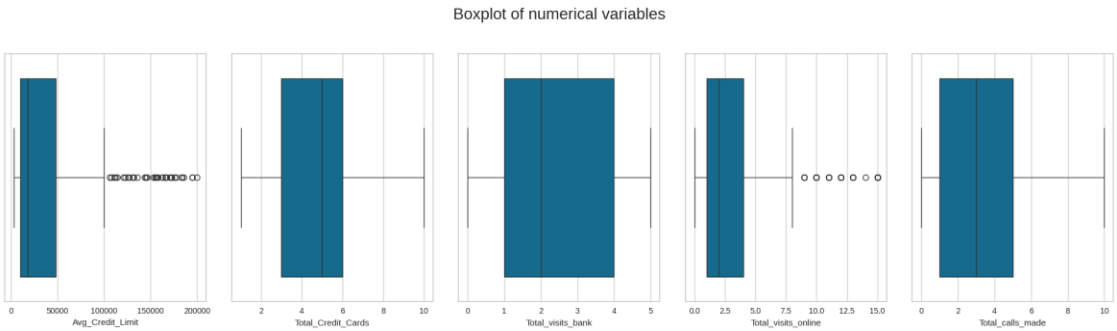
Histogram of numeric values



Observations:

- There are more customers who have 4 to 7 credit cards.
- 67.18% of customers made atleast 2 visits to the bank.
- Most of the customers have made atleast 2 visits online.
- Most of the customers have made 4 calls to the bank.

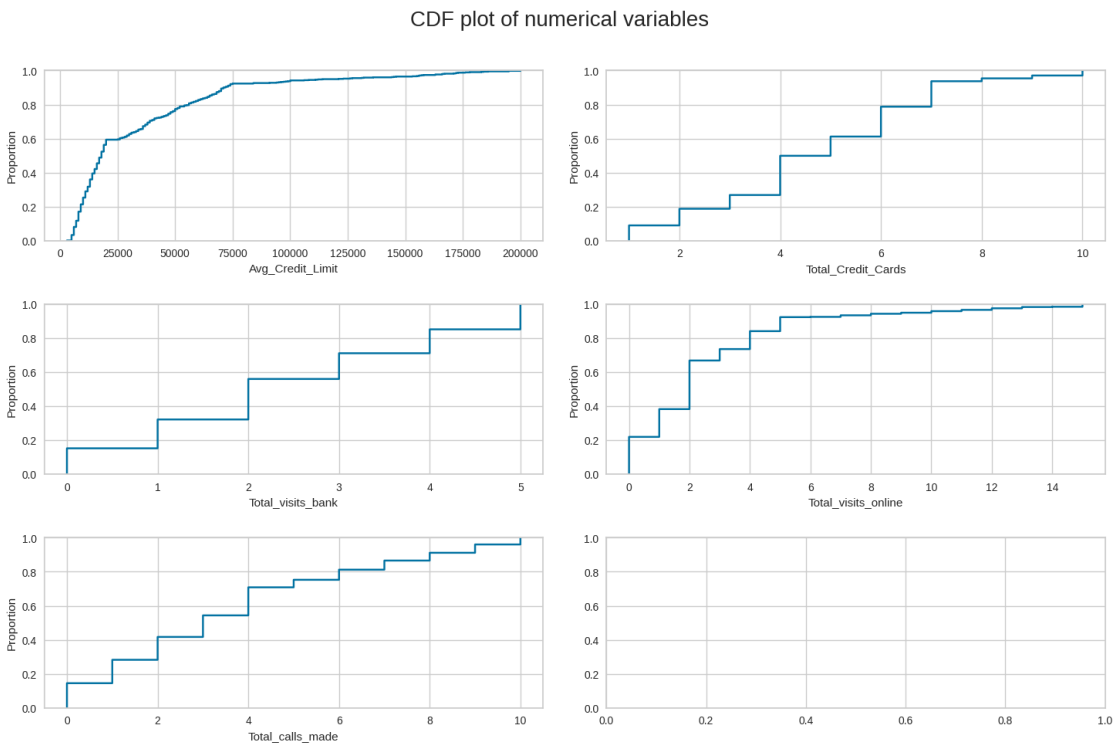
Boxplot of numerical variables



Observations:

- There are upper ouliers in the Avg_Credit_Limit which indicates that the credit limit of few customers is very high.
- 50% of the customers have Avg_Credit_Limit lesser than 18000 while the max is 200000.
- 50% of the customers have atleast 3-6 credit cards.
- 50% of the customers have made 2 visits to the bank.
- There are upper ouliers in the Total_visits_online which indicates that those customers preferred mode of querying the bank is through online, for various reasons. Probably the customers didnt get the satisfactory respone and that is why they are making multiple calls to the bank's customer support.
- 75% of the customers have made upto 4 visits online while the max is 15 visits.
- 75% of the customers have made 5 calls and the max is 10 calls.

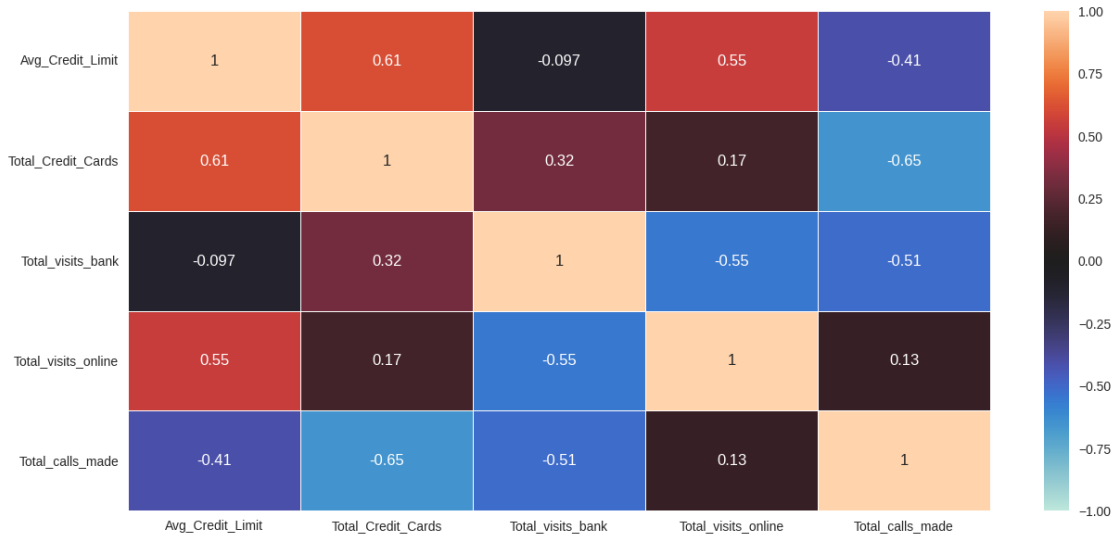
CDF plot of numerical variables



Observations:

- We notice that around 90% of the customers have average credit limit below 75000.
- 60% of the customers have atmost 6 credit cards.
- 90% of the customers have made 5 visits online.
- 80% of the customers have made 6 calls to the bank for enquiries.
- 50% of the customers have made 2 visits to the bank.

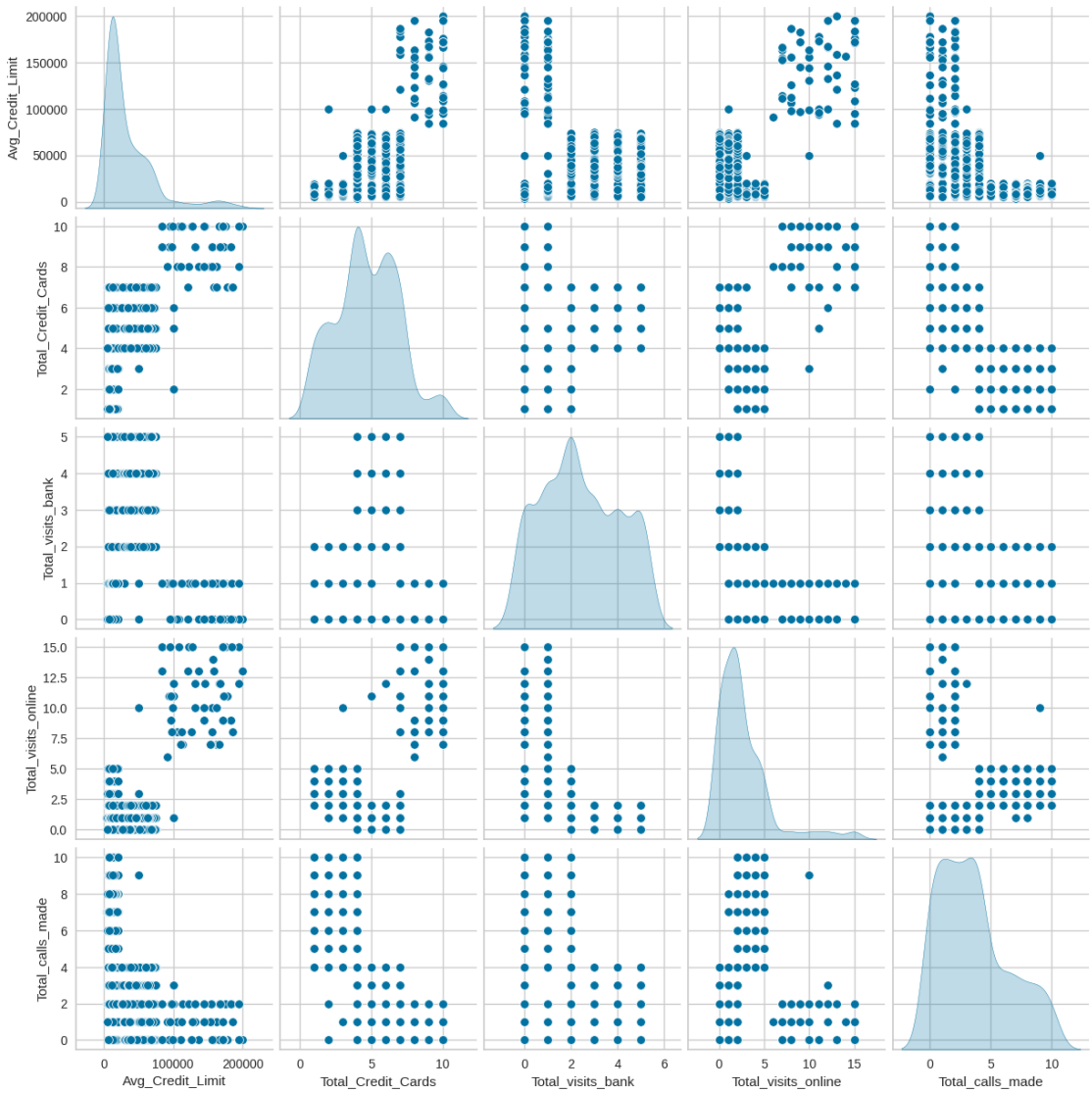
Bivariate Analsysis



Observations:

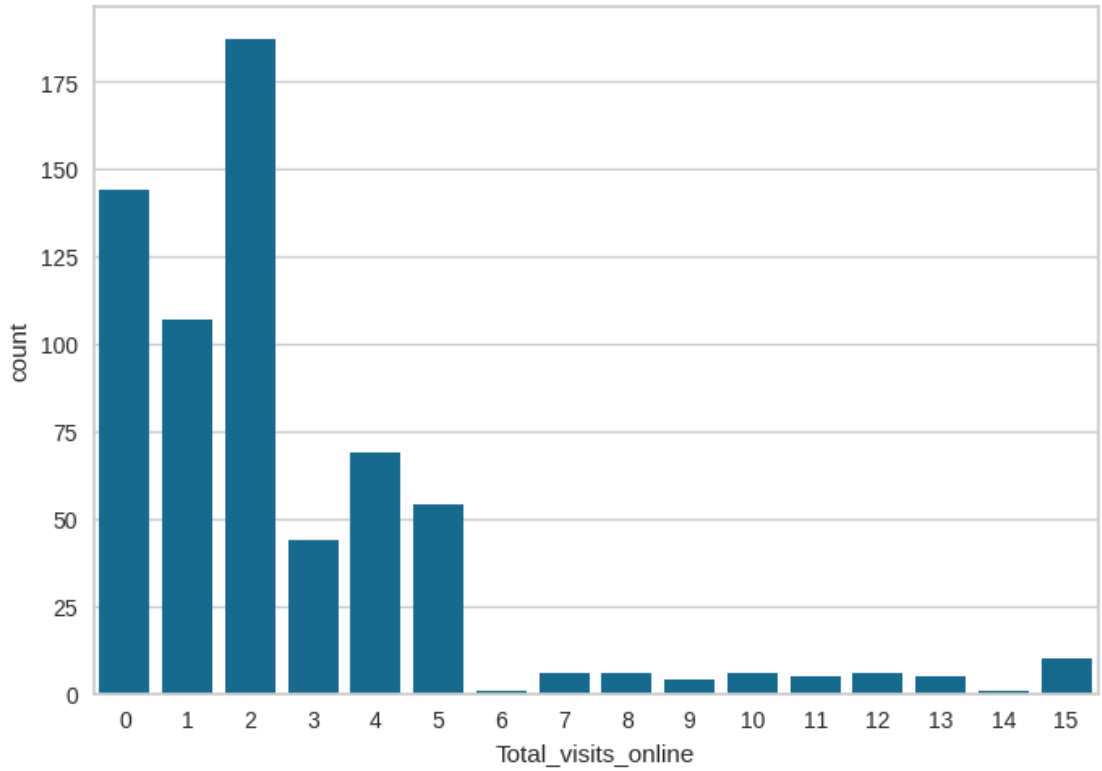
- Avg_Credit_Limit has strong positive correlation with Total_Credit_Cards and Total_Visits_Online and negative correlation with Total_calls_made.
- Total_Credit_Cards has strong positive correlation with Avg_Credit_Limit and negative correlation with Total_calls_made.
- Total_Visits_bank has strong negative correlation with Total_calls_made and Total_Visits_Online

Pairplot of numerical values



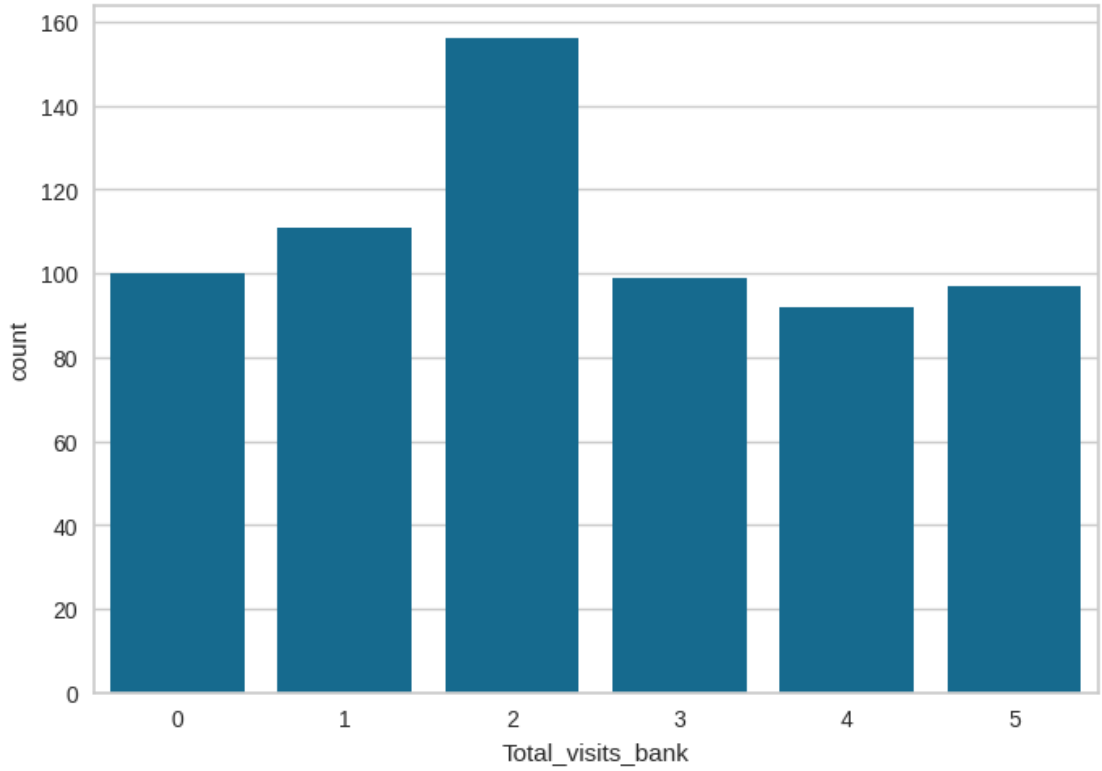
- We notice that Avg_Credit_Limit is showing a linear relationship with Total_Credit_Cards and Total_visits_online.
- From the pair plot, we see that there are atleast 2 peaks for some of the variables, which indicates that there would be a minimum of 2 clsuters.

Count plots of total visits online, total visits bank and total calls made



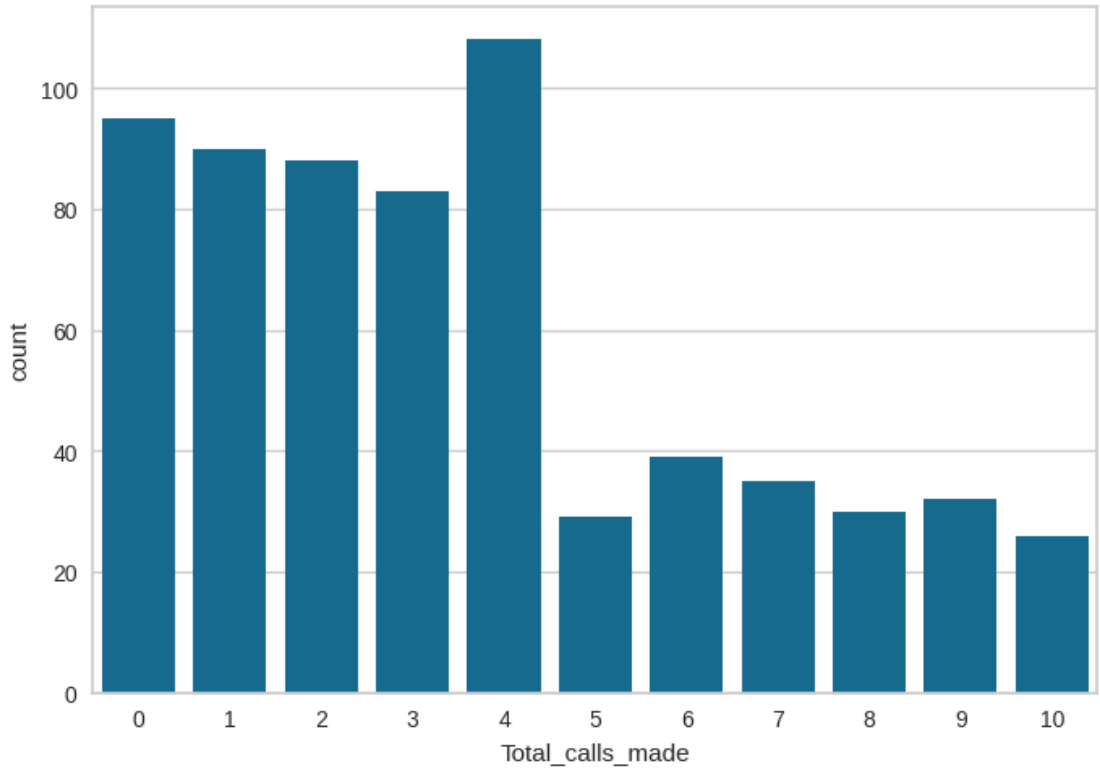
Observations:

- There are more customers(28.8%) who made 2 visits online for queries.
- There are atleast 20% of the customers who donot prefer to make online enqueries.



Observations:

- There are more customers(24%) who visit the bank twice.
- There are atleast 15% of the customers who donot prefer to visit the bank for enquiries.

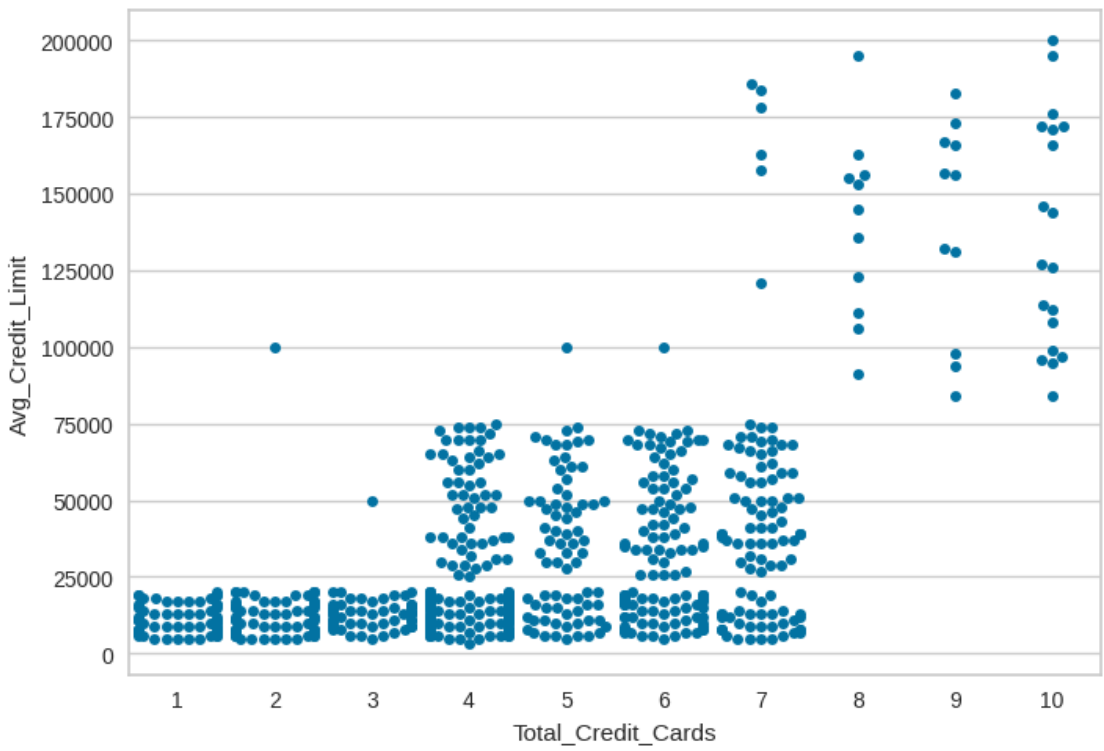


Observations:

- There are more customers(16%) who made 4 calls to the bank for queries.
- There are atleast 14.8% of the customers who donot prefer to make calls to the bank for enquiries.

Swarn plot of Total credit cards and Average credit limit

```
Total_Credit_Cards
1       11457.627119
2       13187.500000
3       13301.886792
4       26311.258278
5       34689.189189
6       32686.956522
7       43163.265306
8      139454.545455
9      140090.909091
10     136842.105263
Name: Avg_Credit_Limit, dtype: float64
```

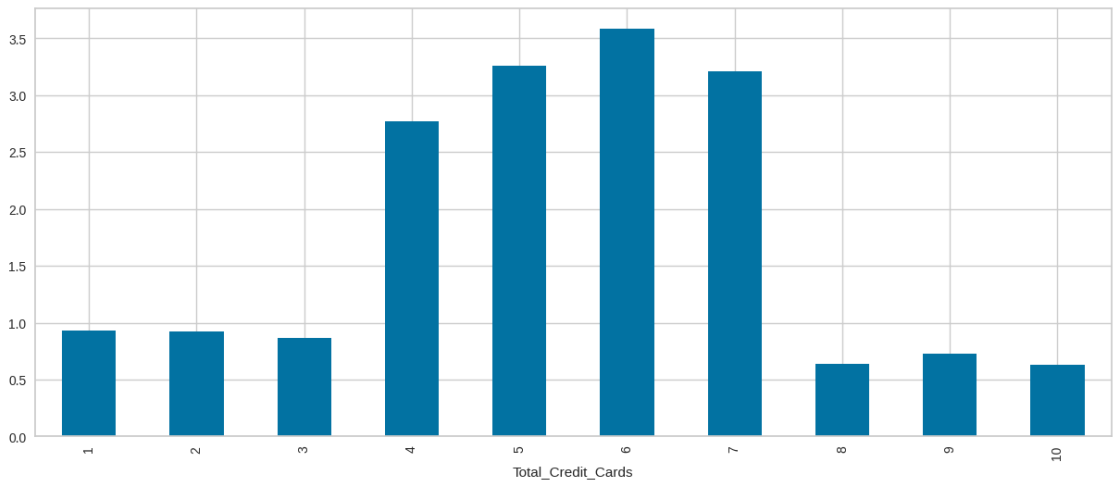


Observations:

- We notoice a positive correlation between Total_Credit_Cards and Avg_Credit_Limit.
- Greater the number of cards, the higher the credit limit.

Barplot of total credit card and total visits bank

```
Total_Credit_Cards
1     0.932203
2     0.921875
3     0.867925
4     2.768212
5     3.256757
6     3.582609
7     3.214286
8     0.636364
9     0.727273
10    0.631579
Name: Total_visits_bank, dtype: float64
```

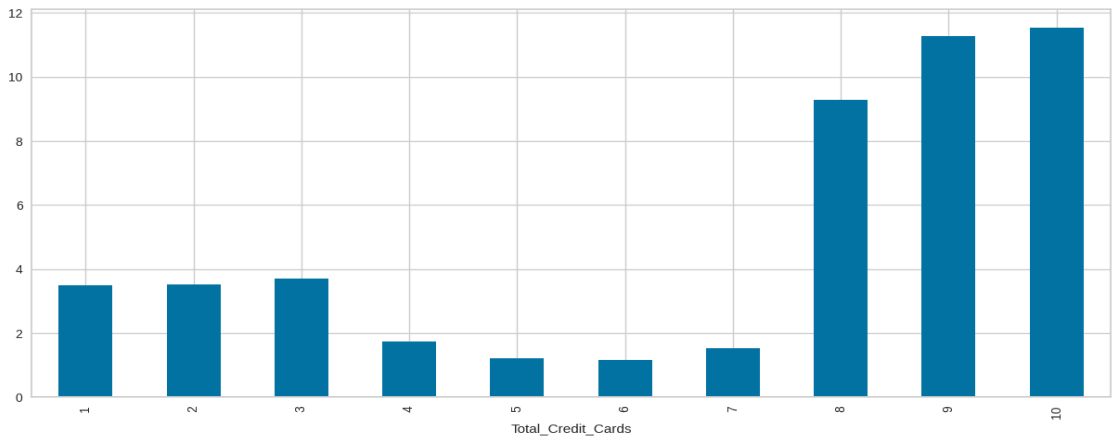


Observations:

- Customers with 4-7 cards have made more visits to the bank, followed by the customers who owned 1 – 3 cards.
- Customers with 8-10 cards made the least number of personal visits to the bank.

Bar plot of total credit cards and total visits online

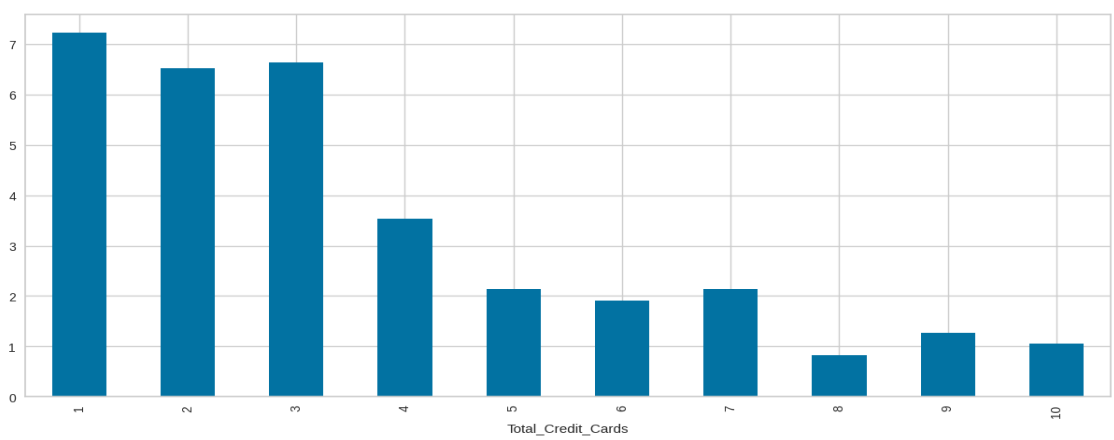
```
Total_Credit_Cards
1      3.491525
2      3.531250
3      3.698113
4      1.728477
5      1.216216
6      1.156522
7      1.530612
8      9.272727
9     11.272727
10    11.526316
Name: Total_visits_online, dtype: float64
```



Observations:

- Customers with 8-10 cards have made more online visits to the bank for enquiries.
- Customers with 1-3 cards made on an average 3 online visits to the bank for enquiries.
- Customers with 4-7 cards have made least number of online visits(one on an average) to the bank.

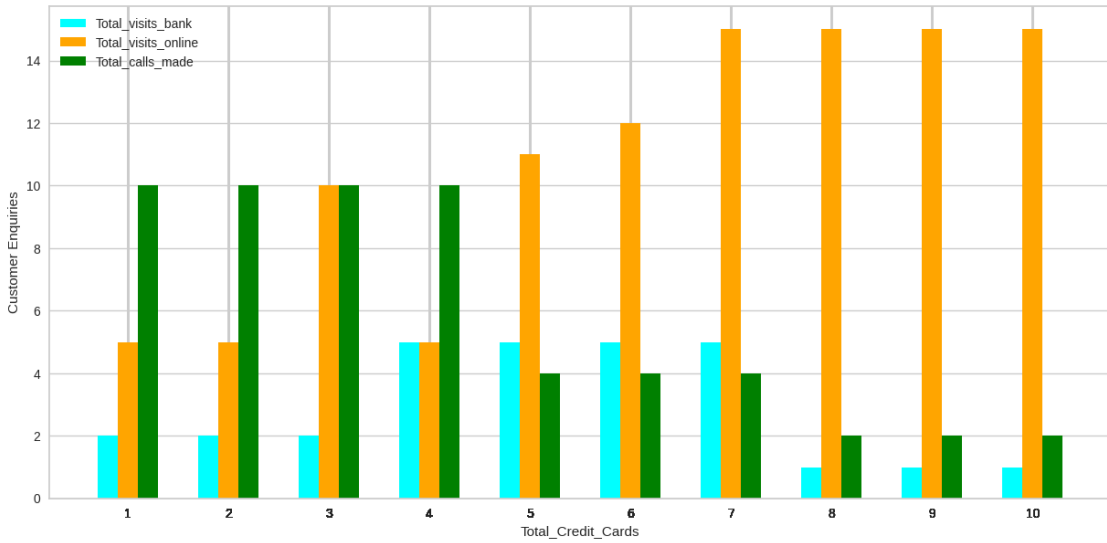
Bar plot of total credit cards and total calls made



Observations:

- Customers with 1-3 cards have made more calls compared to others.

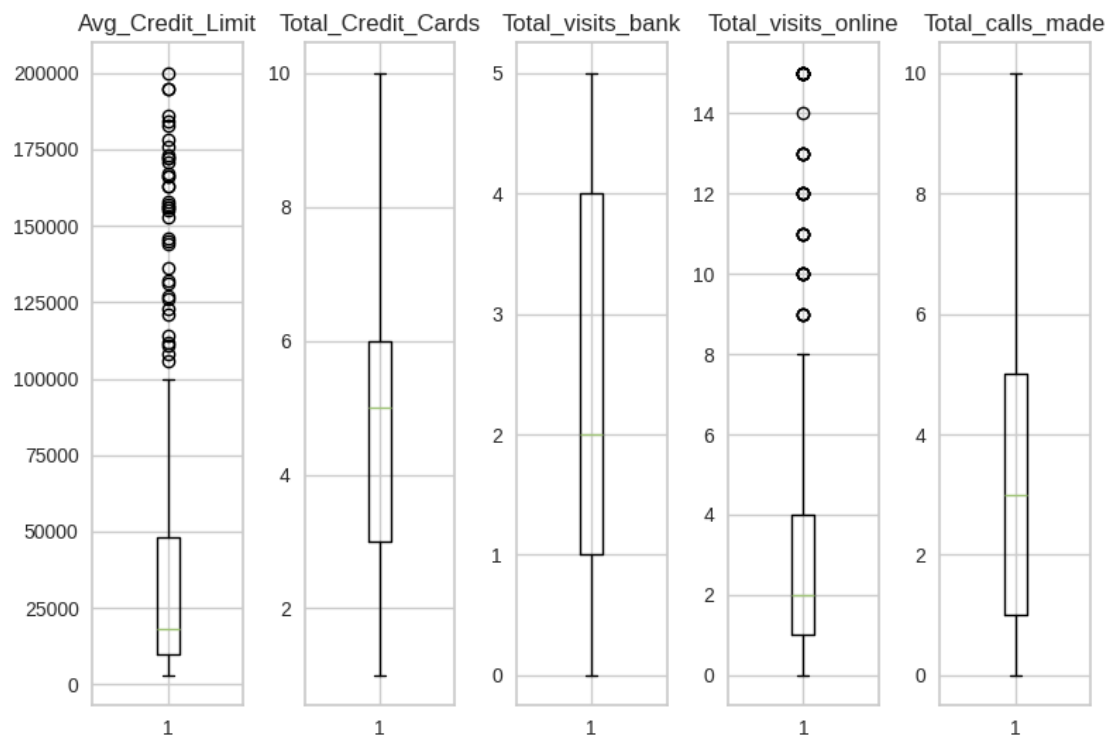
```
# plot data in grouped manner of bar type
```



Treating Outliers

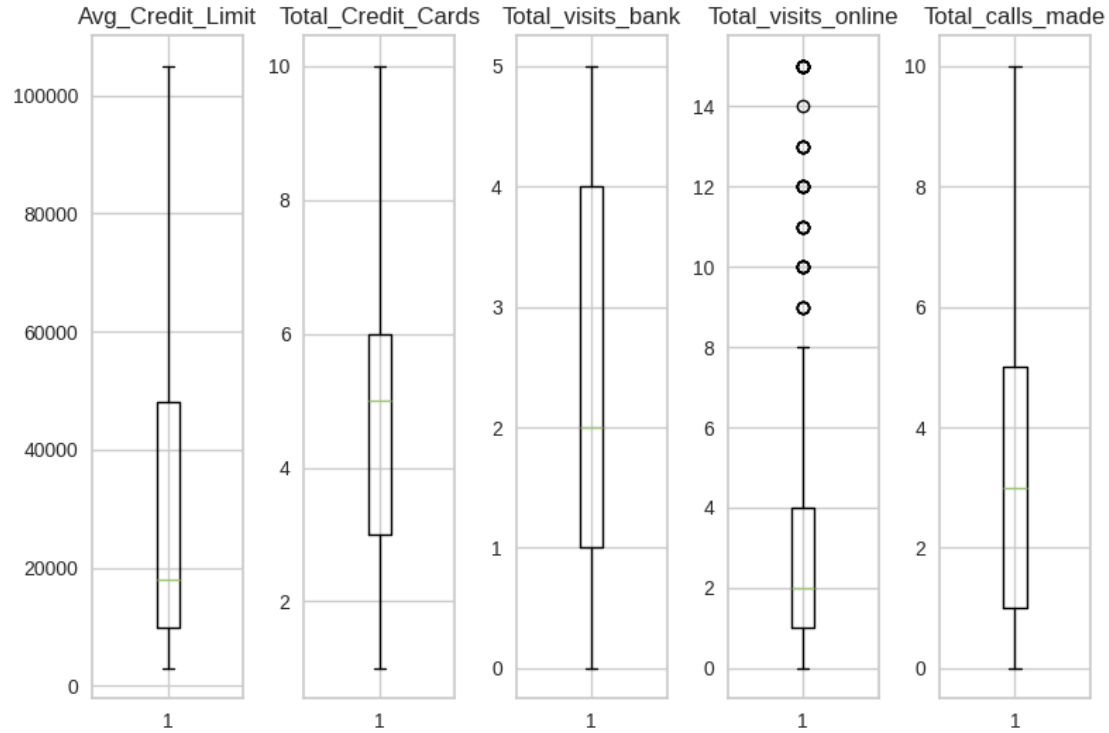
Lets look at outliers in every numerical column

```
# lets look at box plot to see the outliers
```




```
# Lets treat outliers by flooring and capping them at 1.5IQR
# all the values samller than Lower_Whisker will be assigned value of Lower_whisker
# and all the values above upper_whishker will be assigned value of upper_Whisker
```

```
# lets look at box plot to see if outliers has been treated or not
```



- We capped the values at 1.5IQR for only Avg_Credit_Limit column as it is a continuous variable.
- The outliers in Total_visits_online variable are not treated as its values are discrete in nature.

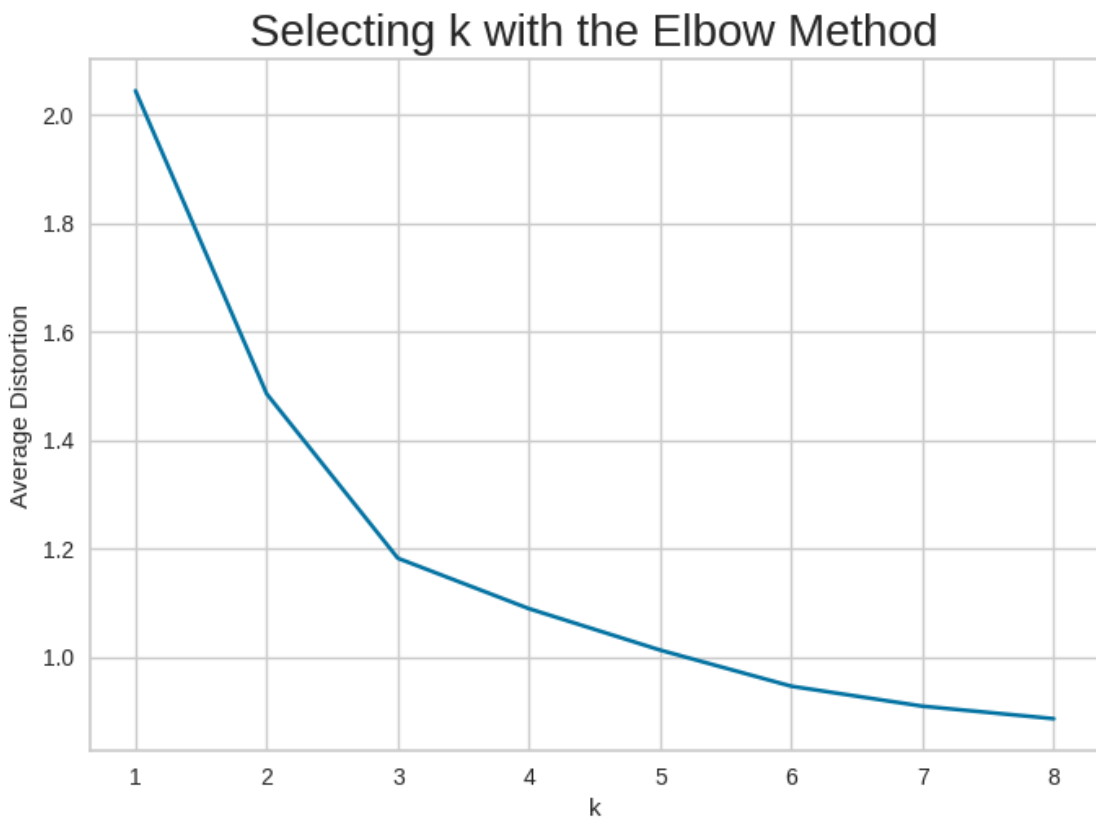
Scaling the data

 # We Scaled the data set before clustering

K-Means Clustering

Selecting K with elbow method

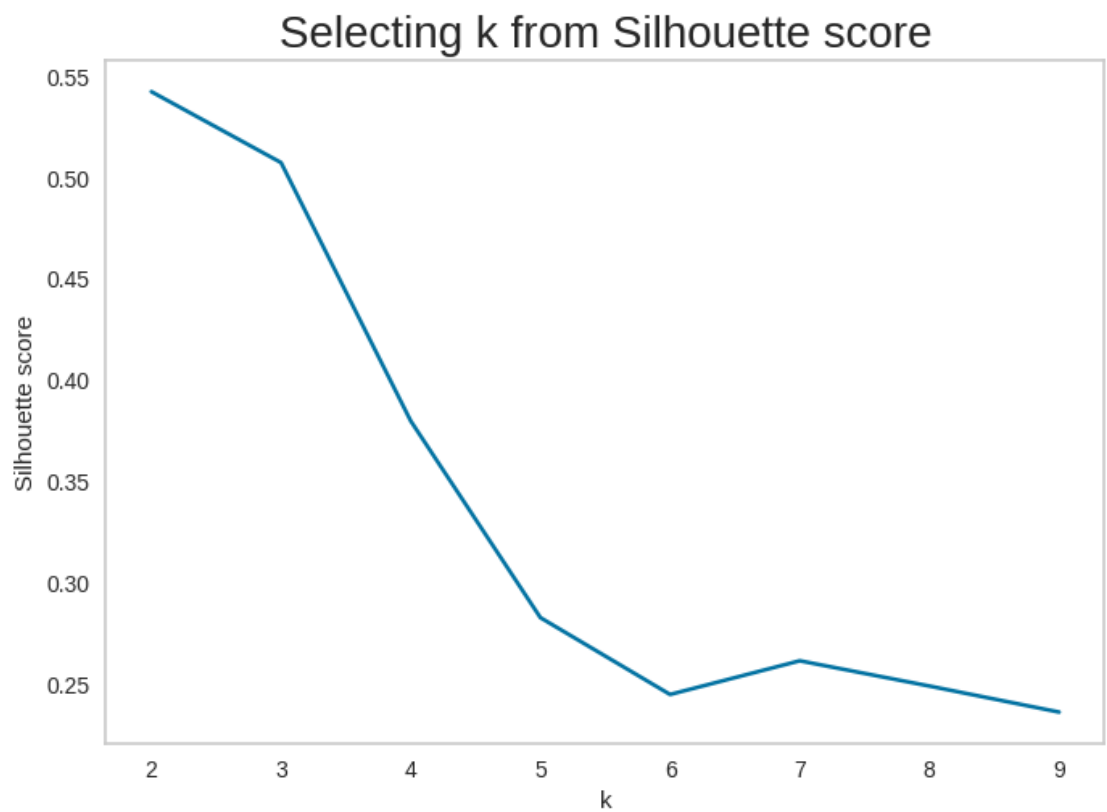
```
Number of Clusters: 1   Average Distortion: 2.044353007149102
Number of Clusters: 2   Average Distortion: 1.4852012536807695
Number of Clusters: 3   Average Distortion: 1.1818148855916517
Number of Clusters: 4   Average Distortion: 1.0887813571635248
Number of Clusters: 5   Average Distortion: 1.0123244952387962
Number of Clusters: 6   Average Distortion: 0.9456753015446184
Number of Clusters: 7   Average Distortion: 0.9088508083166138
Number of Clusters: 8   Average Distortion: 0.8858099551577079
```



From Elbow method, 3 can be a good value for K

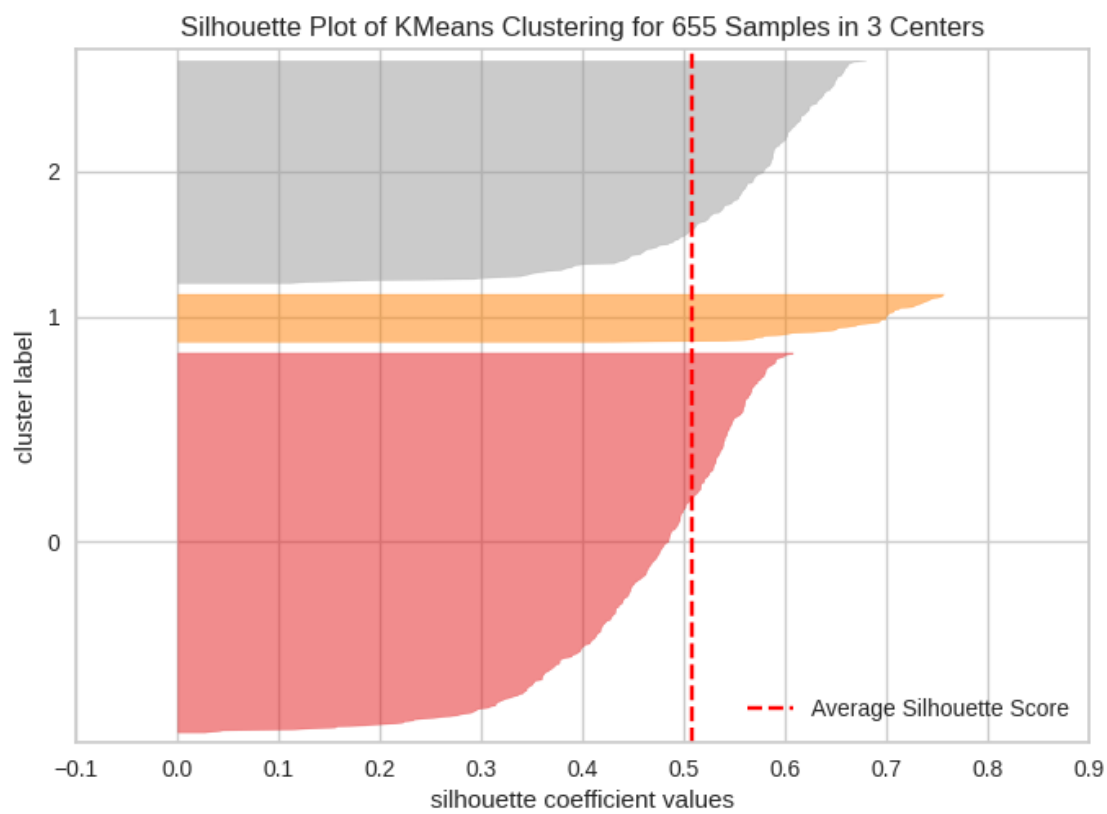
Lets us check silhoutte score

```
For n_clusters = 2, silhouette score is 0.5424748172853491)
For n_clusters = 3, silhouette score is 0.5075035671327102)
For n_clusters = 4, silhouette score is 0.38018134427520495)
For n_clusters = 5, silhouette score is 0.28320876094931935)
For n_clusters = 6, silhouette score is 0.24533850167855498)
For n_clusters = 7, silhouette score is 0.26194073793788686)
For n_clusters = 8, silhouette score is 0.2494897509722565)
For n_clusters = 9, silhouette score is 0.23664824787120436)
```



From silhouette score, it seems that 3 is good value of k

```
#Finding optimal no. of clusters with silhouette coefficients
```



```
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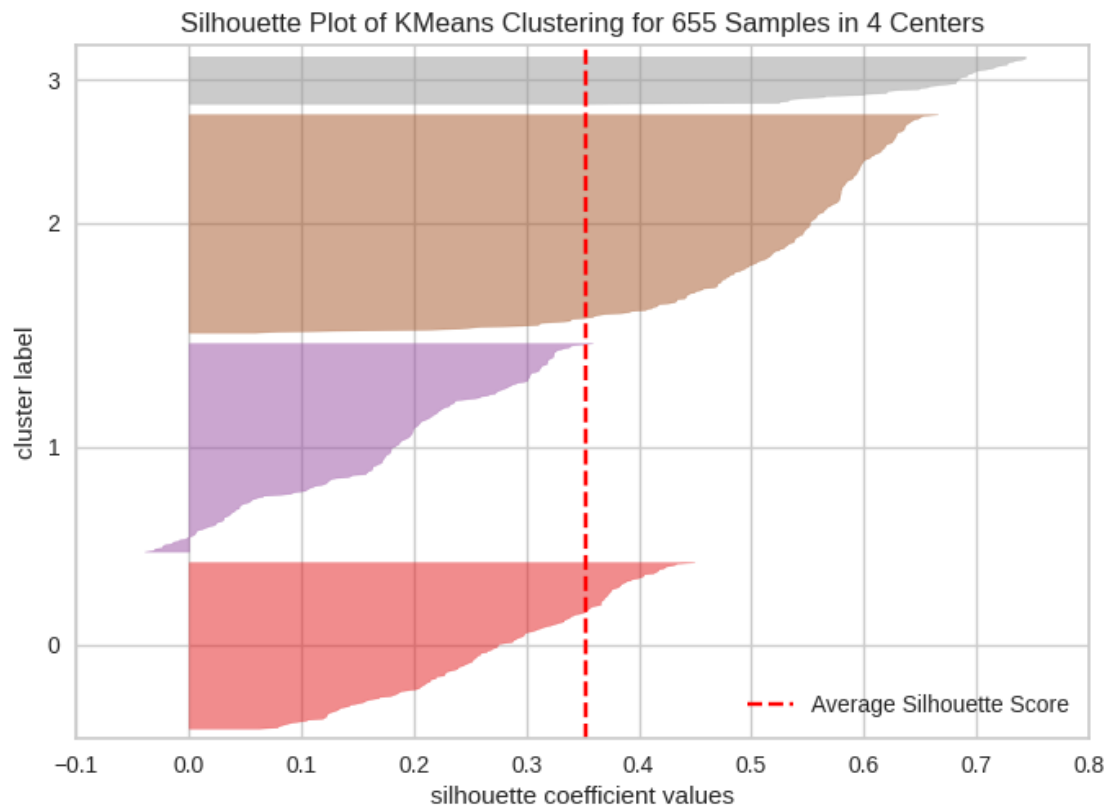


```
#Finding optimal no. of clusters with silhouette coefficients
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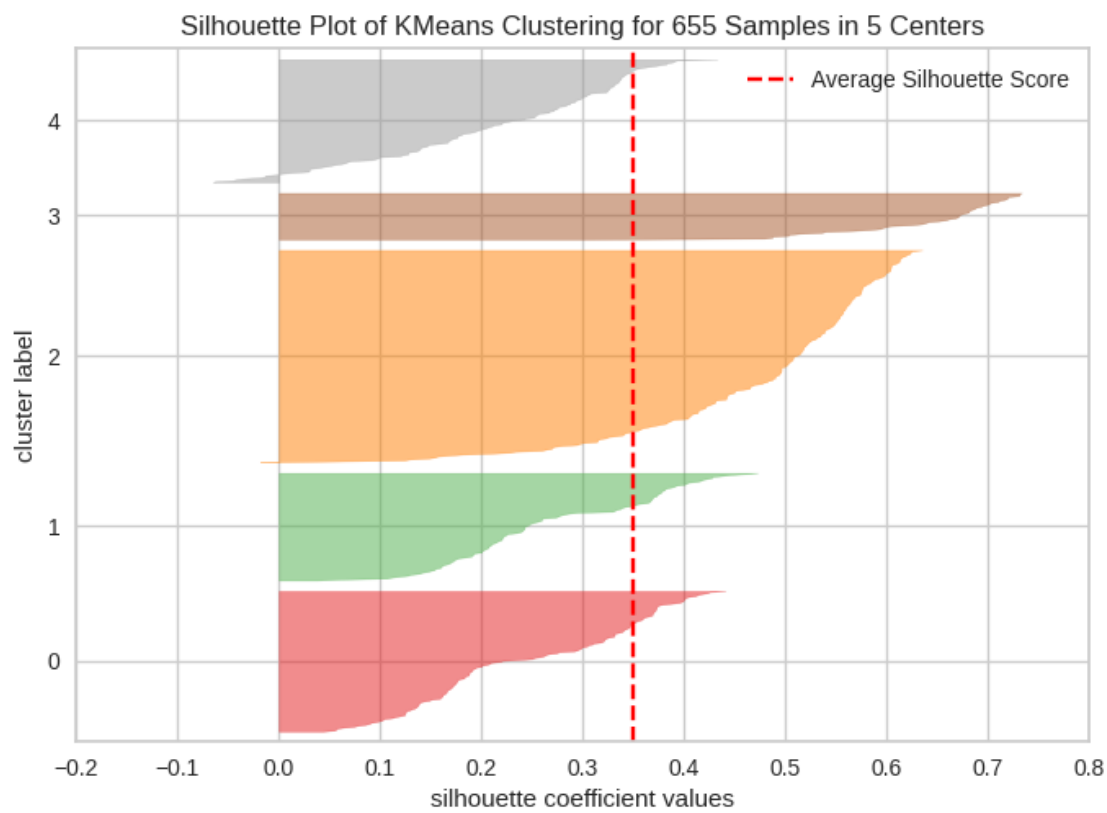

```
#Finding optimal no. of clusters with silhouette coefficients
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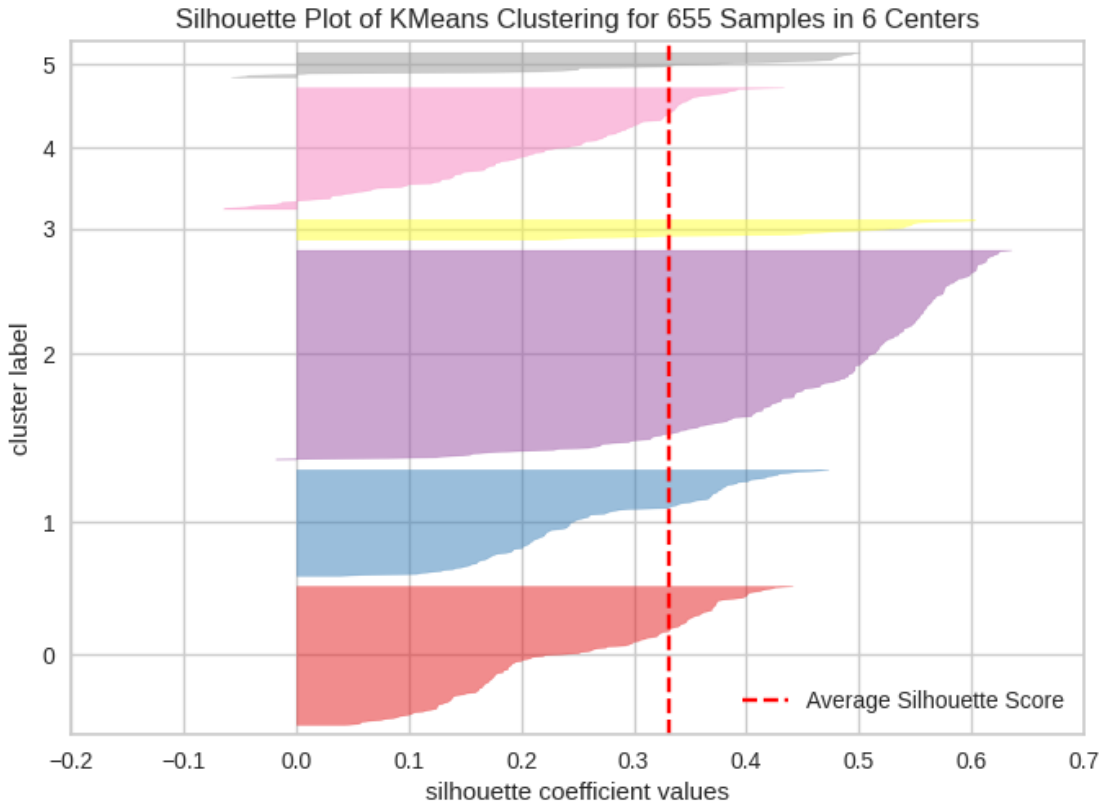
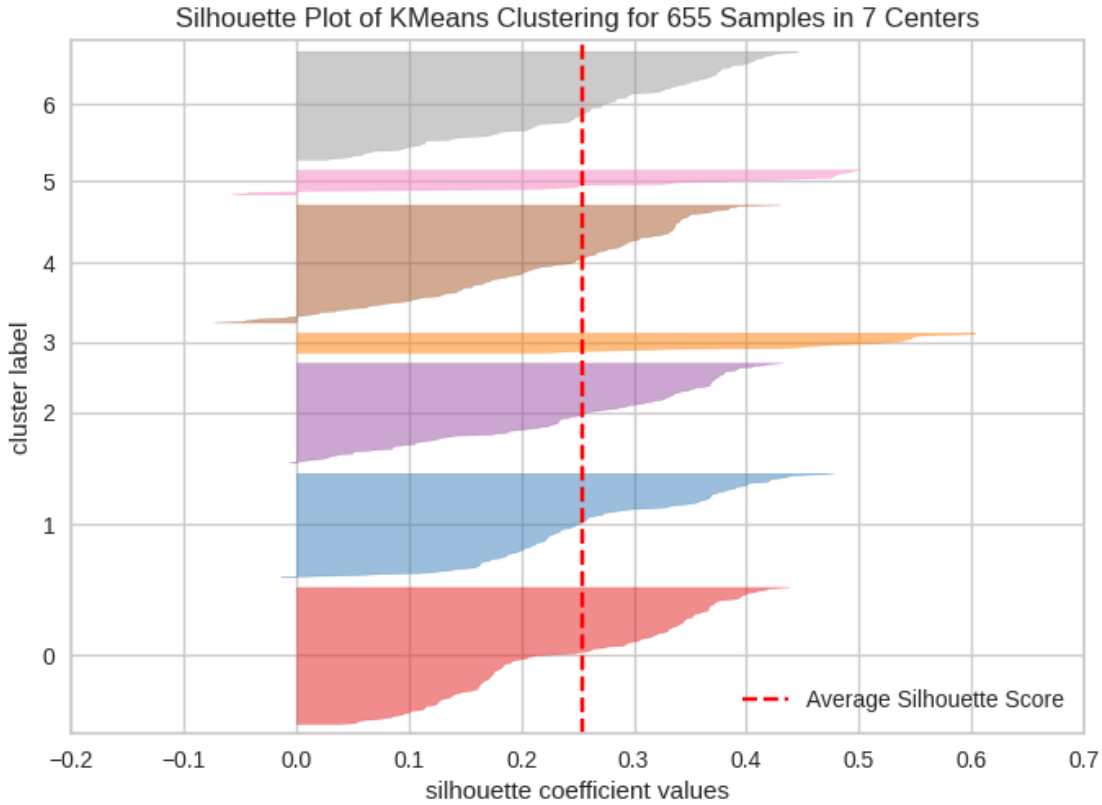
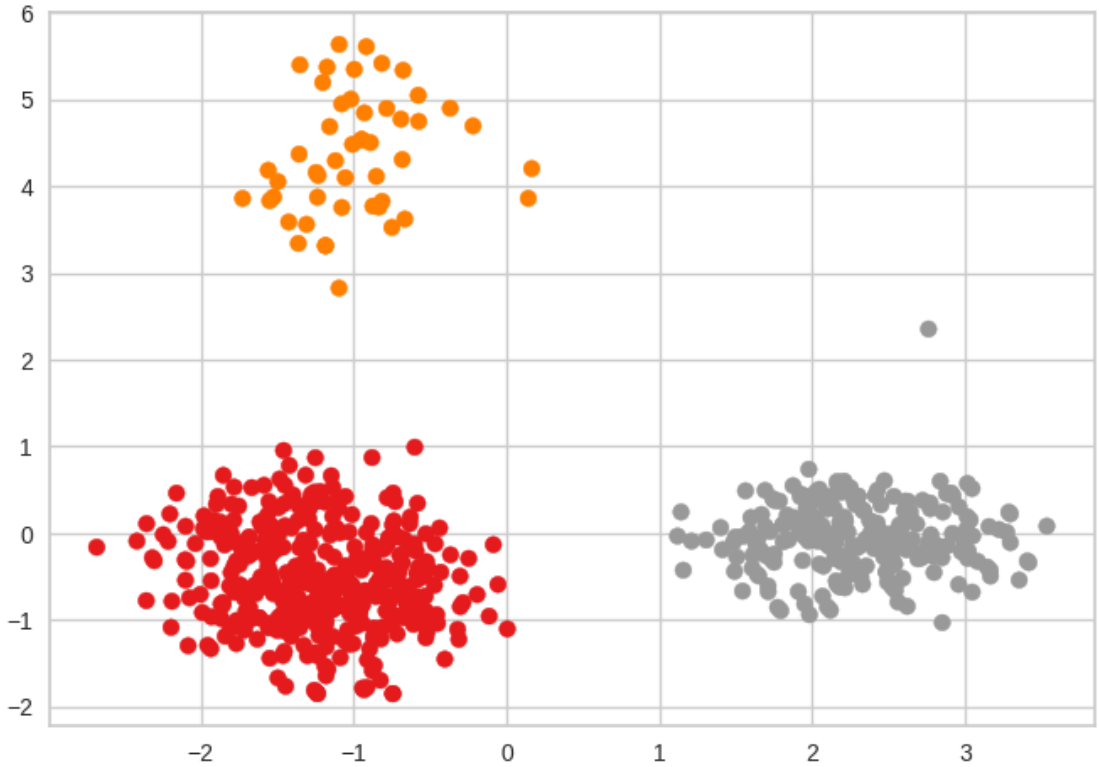
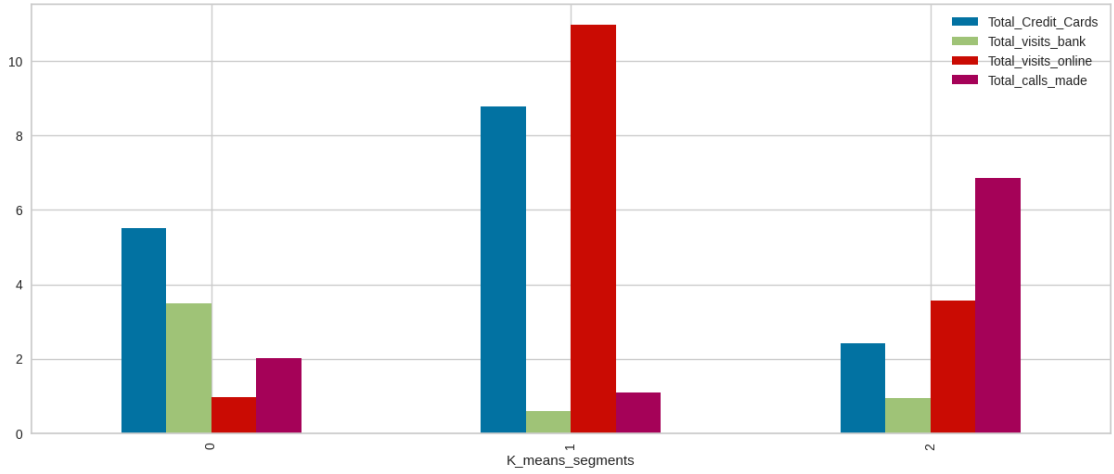


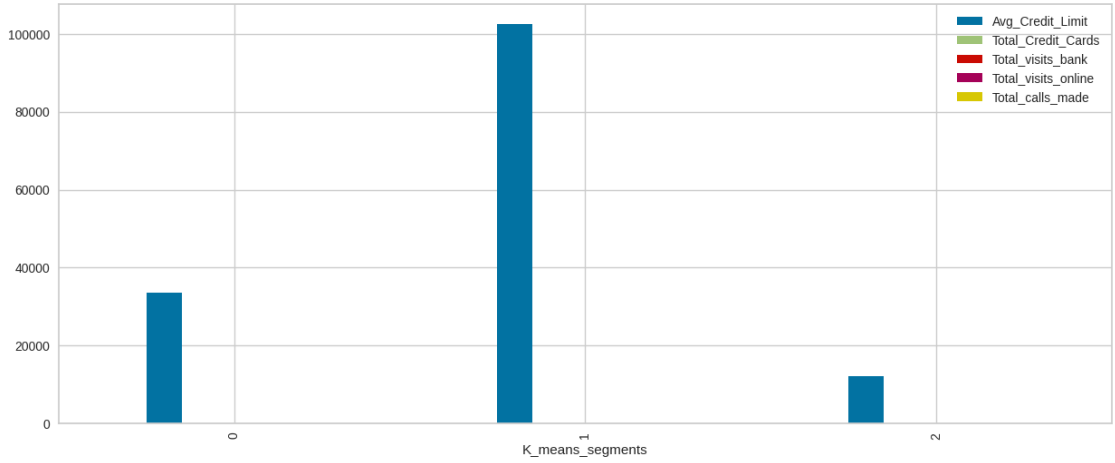

```
#Finding optimal no. of clusters with silhouette coefficients
```

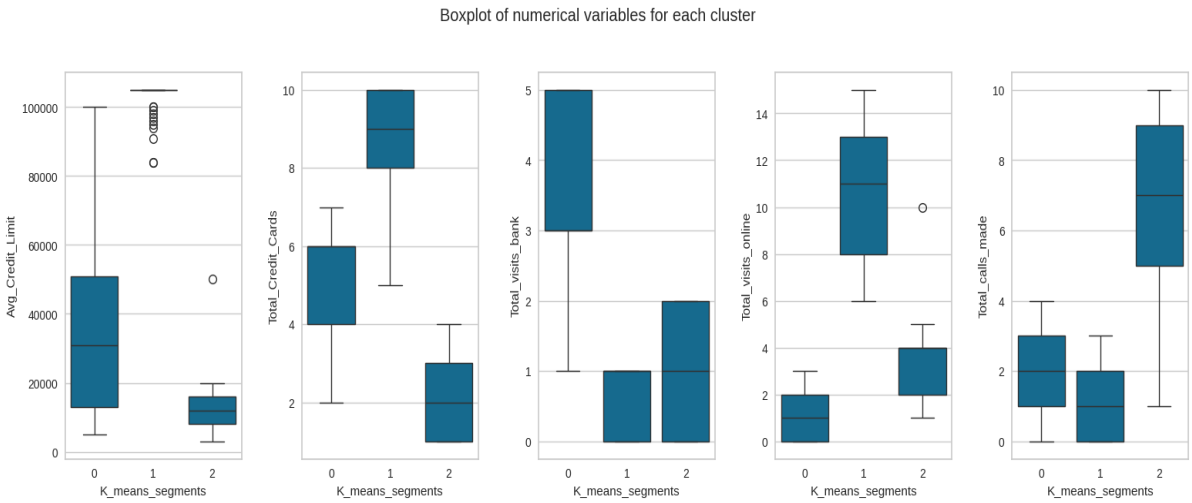


Let us take 3 as appropriate no. of clusters as silhoutte score is high enough and also there is a knick at 3 in elbow curve.









Compare Cluster vs Total_Credit_Cards

Insights

Cluster 0 (Platinum Customers) :

- Avg Credit limit is high for this cluster.
- Total credit cards taken by customers on an average very high.
- Total Visits to the bank is very very low.
- Total Visits online is highest, on an average 11 visits.
- Total calls made is lowest.

Cluster 1 (Silver Customers) :

- Avg Credit limit is low for this cluster.
- Total credit cards taken by customers is very low.
- Total Visits to the bank is low.
- Total Visits online is medium, on an average 3 visits.
- Total calls made is very high compared to other clusters.

Clusters 2 (Gold Customers):

- Avg Credit limit is medium for this cluster
- Total credit cards taken by customers on an average is medium.
- Total Visits to the bank is very high.
- Total Visits online is lowest as minimum as 1.
- Total calls made is lower.

Hierarchial Clustering

```
#cophenetic index is a measure of the correlation between the distance of points in featu
re space and distance on dendrogram
# closer it is to 1, the better is the clustering
```

```
Cophent correlation for distance metrics euclidean and linkage method single is 0.705359467615623
Cophent correlation for distance metrics euclidean and linkage method complete is 0.8668293924257054
Cophent correlation for distance metrics euclidean and linkage method average is 0.8902858447098364
Cophent correlation for distance metrics euclidean and linkage method weighted is 0.8753252634903654
Cophent correlation for distance metrics chebyshev and linkage method single is 0.7219778923794835
Cophent correlation for distance metrics chebyshev and linkage method complete is 0.8388571033820141
Cophent correlation for distance metrics chebyshev and linkage method average is 0.8878999292279559
Cophent correlation for distance metrics chebyshev and linkage method weighted is 0.8189672840833921
Cophent correlation for distance metrics mahalanobis and linkage method single is 0.6770235743969137
Cophent correlation for distance metrics mahalanobis and linkage method complete is 0.6006993693802337
Cophent correlation for distance metrics mahalanobis and linkage method average is 0.8125618164236034
Cophent correlation for distance metrics mahalanobis and linkage method weighted is 0.6986122685173339
Cophent correlation for distance metrics cityblock and linkage method single is 0.6958647046671217
Cophent correlation for distance metrics cityblock and linkage method complete is 0.8426772566439003
Cophent correlation for distance metrics cityblock and linkage method average is 0.8894329744104157
Cophent correlation for distance metrics cityblock and linkage method weighted is 0.8199060509368357
```

```
Highest cophenetic correlation is 0.8902858447098364, which is obtained with euclidean

distance metric and average linkage method
```

Lets explore 'centroid', 'median','ward' linkage method which requires only Euclidean distance metrics

```
Cophent correlation for distance metrics euclidean and linkage method single is 0.705359467615623
Cophent correlation for distance metrics euclidean and linkage method complete is 0.8668293924257054
Cophent correlation for distance metrics euclidean and linkage method average is 0.8902858447098364
Cophent correlation for distance metrics euclidean and linkage method median is 0.8652411270613015
Cophent correlation for distance metrics euclidean and linkage method ward is 0.7433780278360151
Cophent correlation for distance metrics euclidean and linkage method weighted is 0.8753252634903654
*******************************************************************************************
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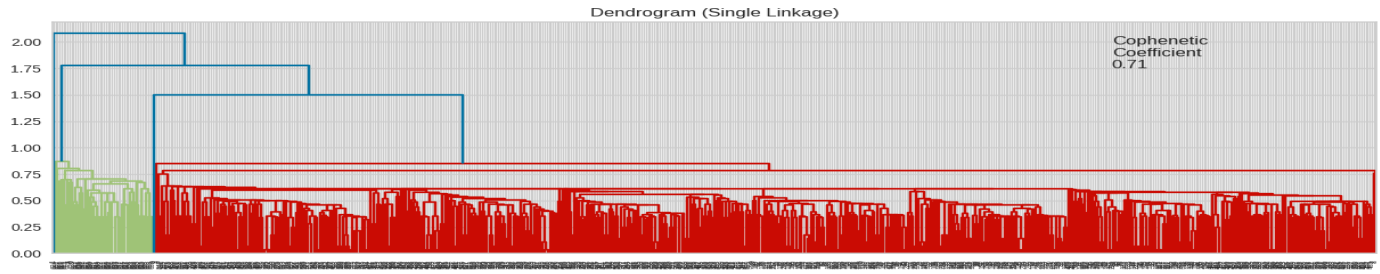
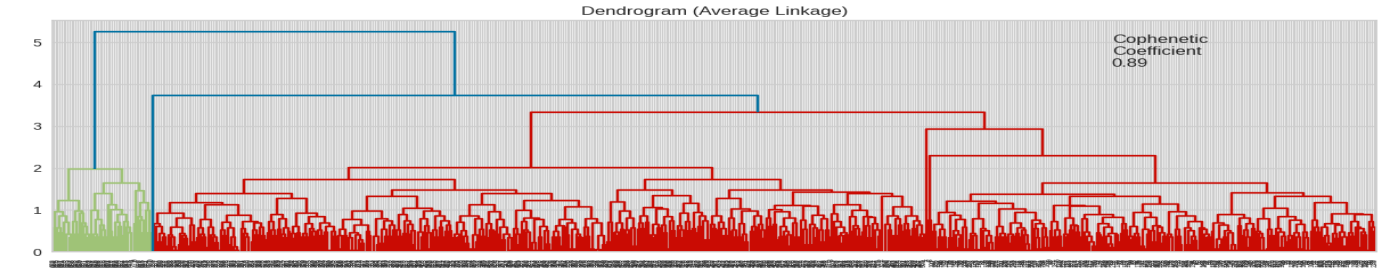
```

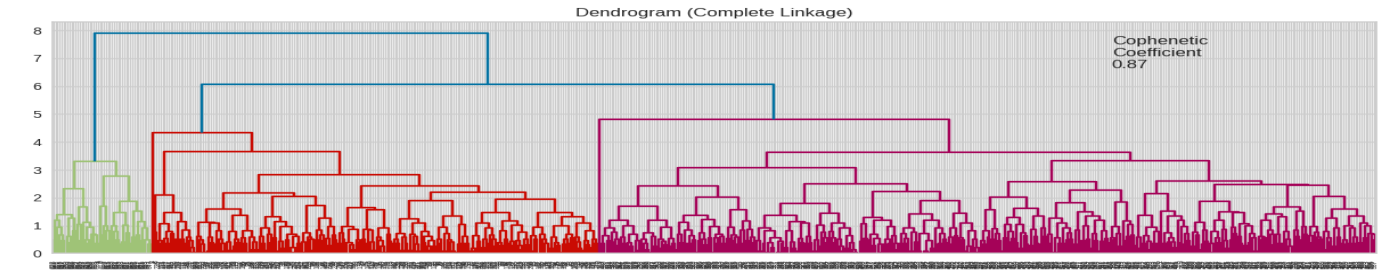
```
Highest cophenetic correlation is 0.8902858447098364, which is obtained with Euclidean di
stance metric and average linkage method
```

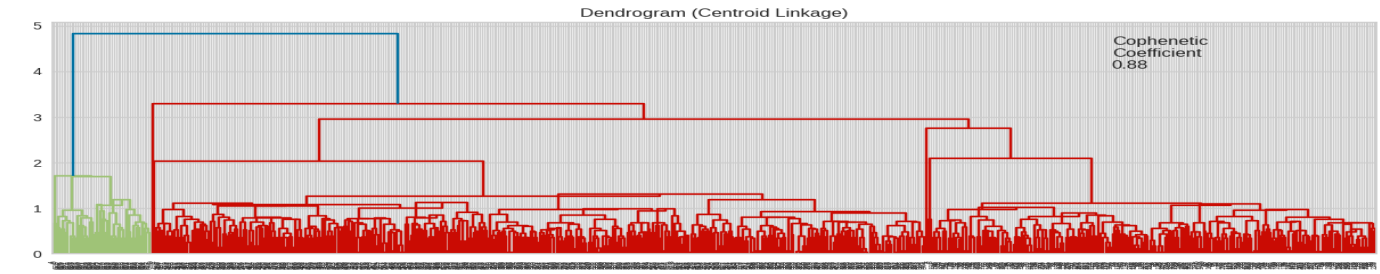
Observations:

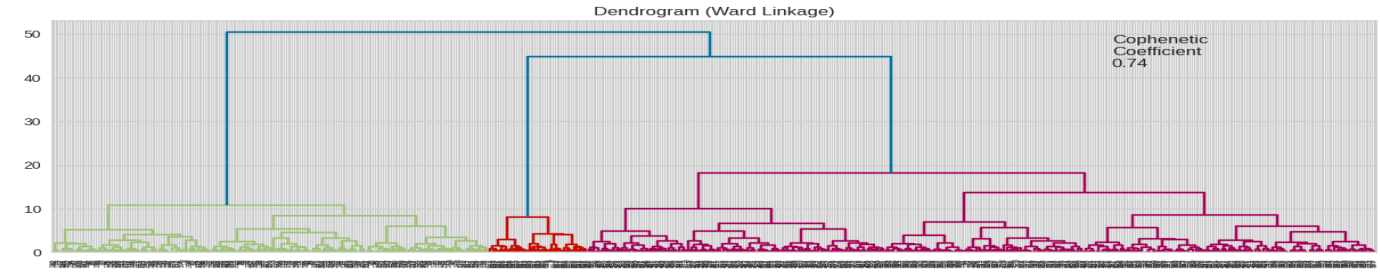
- we see that cophenetic index is maximum with Euclidean distance metric which Euclidean distance method and average linkage method. Lets see the dendrogram of different linkage method

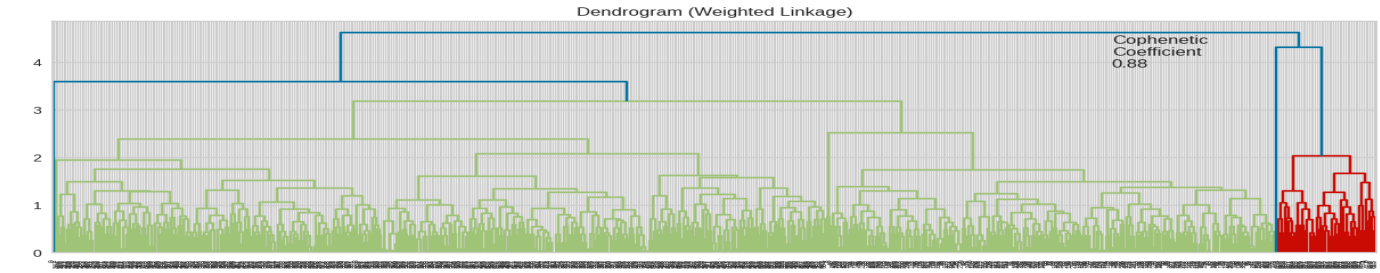






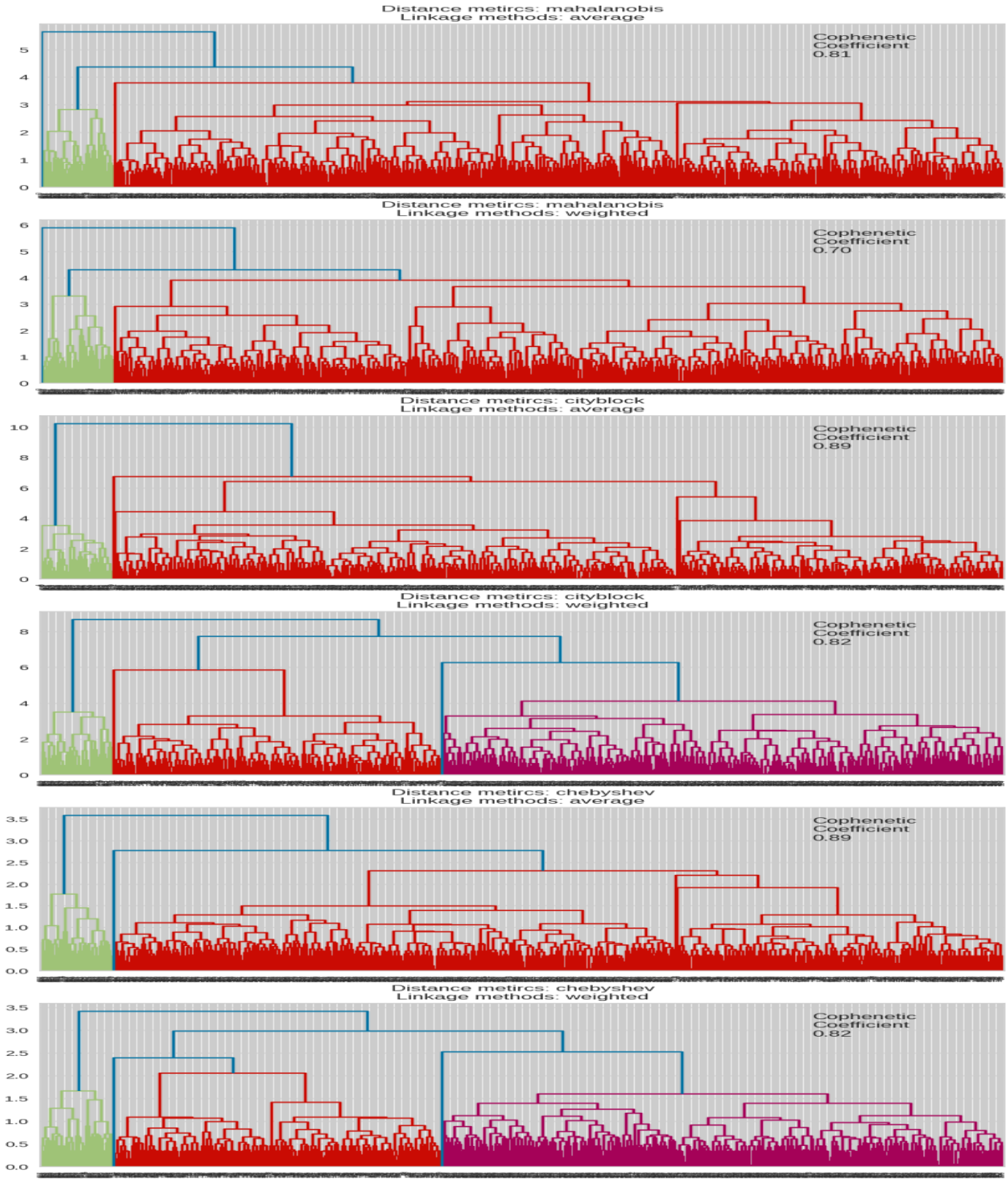






Observations:

- Dendrogram with single and ward linkage method are showing distinct and separate clusters.

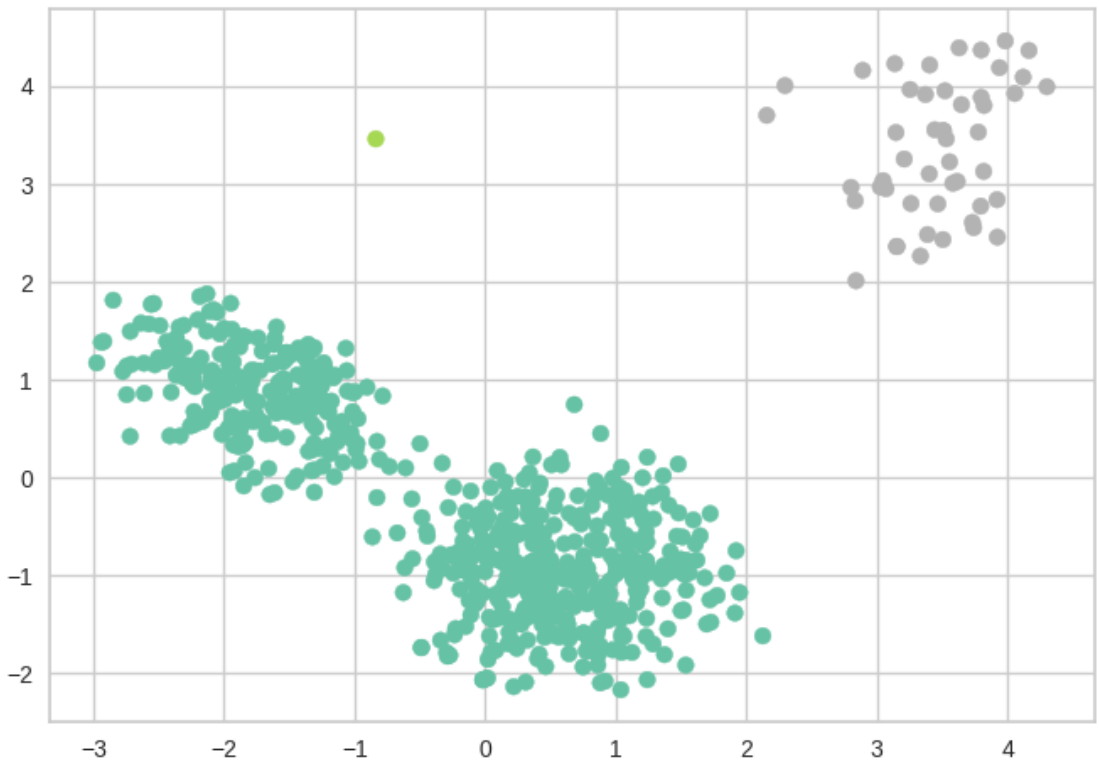


Observations:

- Out of all the dendrogram we saw, it is clear that dendrogram with weighted linkage method gave us separate and distinct clusters
- 3 cluster would be appropriate number of cluster from dendrogram with weighted linkage method

Let us create 3 clusters

```
#Trying with 3 clusters with euclidean distance and single linkage
```



We notice that there is only one record in one cluster with single linkage method which is not optimal.

Insights from Single Linkage Method:

Cluster 0 (Silver Customers):

- Avg Credit limit is less for this cluster
- Total credit cards taken by customers on an average is medium.
- Total Visits to the bank is high.
- Total Visits online is lowest as minimum as 2.
- Total calls made is medium.

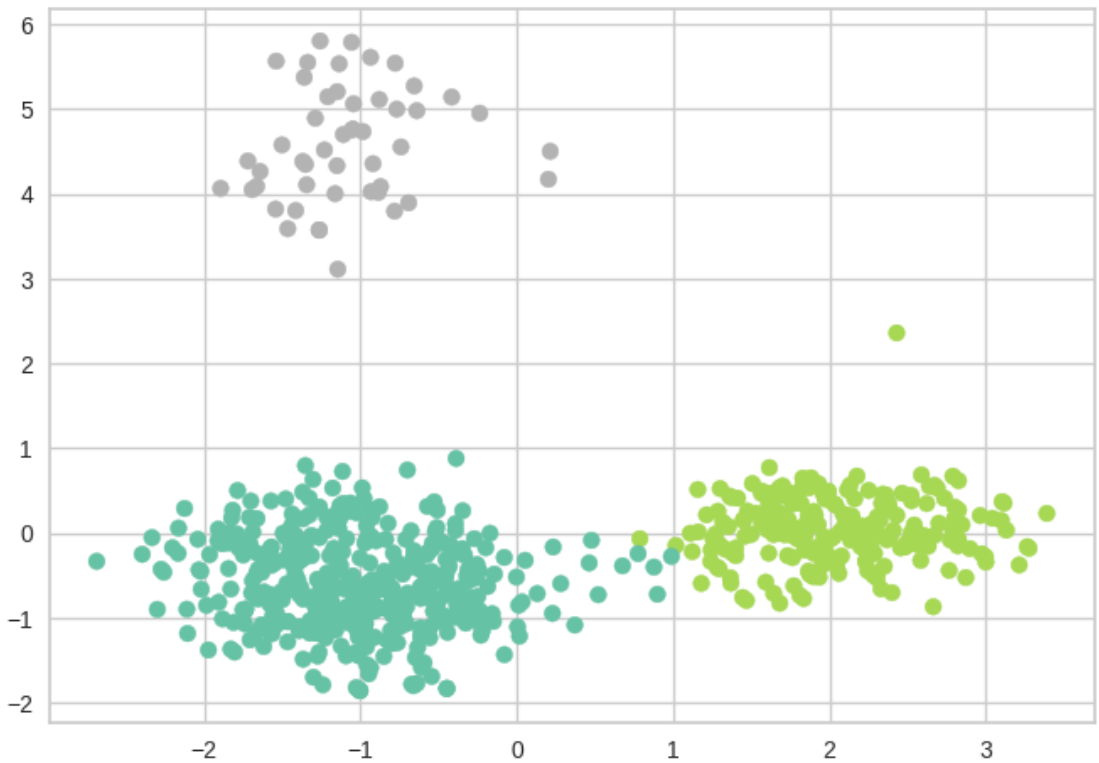
Cluster 1 (Gold Customers):

- Avg Credit limit is medium for this cluster.
- Total credit cards taken by customers on an average is less.
- Total Visits to the bank is zero.
- Total Visits online is high, on an average 10 visits.
- Total calls made is high.

Cluster 2 (Platinum Customers):

- Avg Credit limit is high for this cluster.
- Total credit cards taken by customers on an average very high.
- Total Visits to the bank is very low.
- Total Visits online is highest, on an average 11 visits.
- Total calls made is lowest.
- languages.

```
#Trying with 3 clusters with euclidean distance and ward linkage
# lets display cluster profile
```



We notice that 3 distinct clusters are formed with no overlapping and the results are almost similar to K-Means.

Insights from Ward Linkage Method:

Cluster 0 (Gold Customers):

- Avg Credit limit is medium for this cluster
- Total credit cards taken by customers on an average is medium.
- Total Visits to the bank is high.
- Total Visits online is lowest as minimum as 1.
- Total calls made is medium.

Cluster 1 (Silver Customers):

- Avg Credit limit is low for this cluster.
- Total credit cards taken by customers is less.
- Total Visits to the bank is medium.
- Total Visits online is medium, on an average 3 visits.
- Total calls made is high.

Cluster 2 (Platinum Customers):

- Avg Credit limit is high for this cluster.
- Total credit cards taken by customers on an average very high.
- Total Visits to the bank is very low.
- Total Visits online is highest, on an average 11 visits.
- Total calls made is lowest.

Cluster Comparisions

- K-Means and Hierarchical Clustering- Ward Linkage Methods shows similar results.
- However, Hierarchical Clustering – Single Linkage method gives different set of clusters with different averages and different grouping. The variance in one group is extremely high and there is no uniform distribution and there seems to be an overlap in total visits online from cluster 1 and 2.
- In “Platinum Customer” cluster, 49 customers are identified as platinum customers in all the methods.
- In “Gold Customer” cluster, 382 potential members are identified through K- Means clustering method and 390 members through Hierarchical clustering-Ward Linkage method and 605 customers in Hierarchical clustering-Single Linkage method .
- In “Silver Customer” cluster, 224 members are identified through K- Means clustering method and 216 members through Hierarchical clustering method and 1 customer in Hierarchical clustering-Single Linkage method .
- The differences in Gold and Silver Clusters from both the K-Means and Hierarchical Clustering- Ward Linkage Methods are very close.
- As we know, k-means algorithms assume that clusters are spherical in shapes but that is not the assumption of HC algorithm.
- K-Means clustering gave us a fast solution on k value without much tweaking, though we had to initialize and try on multiple k values.
- With K-Means clustering, both Elbow method and Silhouette score confirmed that k value=3 is apt.
- However, there was some tweaking with Hierarchical clustering to conclude on number of clusters as the single linkage method lead us with one of the clusters with a single record which doesn’t seem to be appropriate as there is an overlap in some values and thereby, we had to choose ward linkage method. With ward linkage method, we had to try both 4 clusters and 3 clusters and then conclude on 3 clusters as there was an overlap between values with 4 clusters.

Therefore K-Means Clustering method worked well for this dataset and should be the method employed for classification of the customer base for credit cards.

Business Recommendations:

Platinum Customers -

Avg Credit limit and Total credit cards taken by customers on an average is very high. These customers prefer online visits over phone call and bank visits.

Recommendation –

- These customers already have more than average number of cards. AllLife bank should not focus on upselling to these customers, but rather focus on improving the customer service to retain them.
- These customers prefer online banking, so AllLife should focus on improving the user userface to improve customer experience. It is also recommended that the bank uses surveys to get feedback from these customers and continually improve the FAQs section to answer all the questions.
- It is also recommended to provide 24x7 online customer support in the form of live chat and emails. AllLife bank can also distinguish itself from other banks by providing a guaranteed response within a short window.
- The platinum customers own a large number of credit cards and can be great ambassadors to the bank to attract newer customers through referrals. AllLife bank should provide incentives to these customers for providing referrals.
- Investing on an app is also a good idea to attract new customers who prefer mobile banking.

Silver Customers -

Avg Credit limit and Total credit cards taken by customers on an average is very low. These customers prefer phone calls over Online and bank visits.

Recommendations-

- These customers have the lowest number of credit cards; there is an opportunity to sell them more by improving their experience with the bank. AllLife bank should, therefore, train the customer service representatives and empower them to resolve the customer issues to reduce the number of phone calls and the time needed to resolve the issues.
- In addition to improving the customer service, AllLife bank can provide additional perks to customers who own greater number of credit cards to encourage existing customers to apply for more credit cards.
- AllLife bank should hire multilingual representatives to cater to customers who speak different languages.

Gold Customers -

Avg Credit limit and Total credit cards taken by customers on an average is medium. These customers prefer bank visits over phone calls and online.

Recommendations-

- The Gold Customers prefer personal visits to the bank. AllLife bank can improve the customer experience by Encouraging customers to schedule appointments to reduce wait times.
- Building a luxury lounge in the bank with amenities (TVs, coffee, internet) for walk in (no appointment) customers while they wait for service.

- Following up with the customers after their visit to check if their issues have been resolved. This provides AllLife representatives an opportunity to address any unresolved issues and improve customer experience and loyalty.